Hands-on Lab: Vision–Language Models

Duration: 50 minutes — Work in pairs

In the lecture we saw how a Vision–Language Model (VLM) connects an image to language. Today you will *run* two of these models yourself, see what they do well, and, most importantly, find where they fail and explain why.

Before We Start

- Each pair needs **one laptop**. One partner runs the notebook; the other reads the manual, suggests what to try next and records results in the worksheet. **Swap roles after Step 2.**
- If there are not enough laptops, a group of three is allowed, but every member's name must appear on the worksheet.
- Open the Colab notebook shared with you. Go to **Runtime → Change runtime type → T4 GPU**.
- **Run Step 0 immediately.** It downloads the models and takes 2–4 minutes. Read Section 1 while it runs.
- You only need to edit lines marked `# EDIT`. You are not expected to write new code.

Table 1: Plan for today's session

Time	Notebook step	What you do
0–5 min	Step 0	Form teams, load the models
5–20 min	Step 1	CLIP: classify images using only text
20–30 min	Step 2	Ask a generative VLM questions
30–40 min	Step 3	Break the VLM and explain the failure
40–50 min	Step 4	Complete and submit the worksheet

1 The Two Models We Will Use

Definition: CLIP (contrastive model)

CLIP has an **image encoder** and a **text encoder**. It was trained with *contrastive learning*: matching image–caption pairs are pulled close together in a shared embedding space, and non-matching pairs are pushed apart. CLIP does not generate text. It only tells you *how similar* an image is to each piece of text you give it.

Definition: SmolVLM (generative VLM)

SmolVLM is a small generative VLM. A **vision encoder** turns the image into a sequence of visual tokens, a **projector** maps those tokens into the language model's embedding space, and a small **LLM** then writes the answer one token at a time, just like a chatbot.

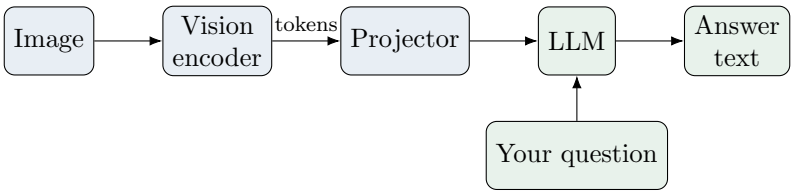


Table 2: CLIP versus a generative VLM

	CLIP	**SmolVLM**
Output	A similarity score per label	Free-form text
You must provide	A list of candidate labels	A question
Good for	Fast classification, retrieval	Captioning, question answering
Typical weakness	Only as good as your labels	Can sound confident and be wrong

2 Step 1 — CLIP: Zero-Shot Classification

CLIP turns the image and each label into embeddings, computes the **cosine similarity** between them, and applies a softmax to get percentages. "Zero-shot" means the model was never trained on your classes: you define the classes just by typing them.

Example

The notebook loads a photo of two cats on a sofa. With labels `["cat", "dog", "car", "sofa"]`, CLIP puts most of the probability on `cat`. Notice that `sofa` also gets a share: the sofa really is in the picture, but softmax forces the labels to compete for 100%.

Try It Yourself

1a. Replace the labels with `"one cat"`, `"two cats"`, `"three cats"`. Does CLIP know how many cats there are?

1b. Change the `prefix`. Compare `"a photo of "`, an empty prefix `""`, and `"a drawing of "`. Record how the top score moves.

1c. Add a silly label such as `"a spaceship"`. What score does it get, and why is it not exactly 0%?

Question

Why can the same image get a different top score when you only change the words *around* the label?

3 Step 2 — Asking a VLM Questions

Now we switch to SmolVLM. Because it generates text, you can ask open questions instead of supplying labels.

Try It Yourself

2a. Ask for a one-sentence caption of the cats image.

2b. Ask two specific questions, for example the colour of the sofa, or what objects lie next to the cats.

2c. Ask the *same* question in two different wordings. Is the answer the same?

4 Step 3 — Break the VLM

This is the main part of the lab. The notebook *draws* test images for you, so you always know the correct answer. Your team's job is to find a clear failure in each category and then **change one**

thing (the prompt or the image) to test your explanation.

Table 3: Failure categories to investigate

Category	What to try	Possible reason to test
Counting	3, 7, 12, 20 circles	The image is cut into patches; no part of the model "counts"
Left vs right	Swap the two shapes	Spatial position can be lost when visual tokens are compressed
Reading text	Small font, rotated text	Low input resolution; text pixels get blurred
Hallucination	Ask about a dog that is not there	The language model's prior overrides the image
Your own photo	Whiteboard, handwriting, crowded desk	Anything above, or something new

Example: a good failure report

(Illustrative numbers.)
Failure: With 12 circles, the model answered "8".
Our explanation: the model sees patches, not individual objects, so it estimates rather than counts.
Test: we tried 3 circles and it answered "3" correctly, and with 20 it answered "10". The error grows with the number of objects, which supports our explanation.

Question

In 3d you ask "What is the dog doing in this picture?" about an image with no dog. Why is this question harder for the model than "Is there a dog in this picture?"

Try It Yourself: your own photo

Take a photo on your phone, send it to the team laptop (WhatsApp Web, email or Google Drive), and upload it in Step 3e. Ask SmolVLM to describe it, then run the CLIP bonus cell on the same photo. Does one model succeed where the other fails?

Questions

Q1. Why does CLIP need a list of labels while SmolVLM does not?
Answer: CLIP only measures similarity between an image embedding and text embeddings, so it can rank options but cannot invent an answer. SmolVLM has a language model that generates text token by token, so it can produce an answer that was never in a list.

Q2. Why do CLIP's percentages always add up to 100%, even if none of the labels is correct?
Answer: The scores are passed through a softmax over the labels you supplied. Softmax spreads the probability across those labels only, so even a poor set of labels will have a "winner".

Q3. Why can changing the prefix from "a photo of" to nothing change CLIP's scores?
Answer: CLIP was trained on real image captions, which usually look like sentences ("a photo of a cat"). A bare word produces a text embedding that is less like those captions, so its cosine similarity with the image changes.

Q4. Give one reason a VLM struggles to count many objects.
Answer: The vision encoder splits the image into patches and produces a fixed number of visual tokens. There is no explicit counting step, so the model estimates the number from an overall impression, which becomes less accurate as the number grows.

Q5. Why might a VLM read large text correctly but fail on small or rotated text?
Answer: The image is resized to the encoder's input resolution. Small text is reduced to only a few pixels per letter, and rotated text is less common in the training data, so letters become hard to recognise.

Q6. What is hallucination in a VLM, and where does it come from?
Answer: Hallucination is describing something that is not in the image. It comes from the language model part: the question ("what is the dog doing?") makes a dog very likely in the text, and this language prior can outweigh the weak visual evidence.

Q7. SmolVLM is a very small model. Would a large commercial VLM make the same mistakes?
Answer: It would make fewer of them, because it uses higher input resolution, a larger language model and more training data. The same failure types still appear in large models, though usually only on harder cases such as many objects or very small text.

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Worksheet (submit one per team)

Team members: _____

Table 4: Step 1 — CLIP results

Labels / prefix tried	**Top label and score**	**Surprised? (Y/N)**

Table 5: Step 3 — Failures found

Category	**Model's answer vs. true answer**	**Our explanation**	**One change we tried, and the result**
Counting			
Left/right			
Text (OCR)			
Hallucination			
Own photo			

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Our best failure (we may show this one on the projector):

_____

_____

_____

In one sentence: when would you use CLIP instead of a generative VLM?

_____

_____